

Macroeconomics III
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Chapter 9

R&D-BASED
ENDOGENOUS
GROWTH:
MACROECONOMIC
MODELLING

INTRODUCTION

- In all of our growth models the ultimate source of long-run growth in income per capita is **technological progress** (growth in A_t).
- In the growth models we have seen so far technological progress has been either:
 - **Unexplained** (exogenous growth: Chapters 5, 6 and 7) or
 - **Unintentional** (endogenous growth model based on productive externalities: Chapter 8).
- In the real world many people are working with **intentional** production of better technology: R&D.
- Now: Endogenous growth models where technological progress is an endogenous and intentional result of an input consuming production process, i.e., growth models with an explicitly formulated **R&D production process**

- The final output of the R&D sector is: **technological progress** $= A_{t+1} - A_t$.
- The direct output of the R&D sector is: **ideas**.
- R&D is a labour-intensive production process! We assume it does not involve capital at all.
- The R&D production process:

Inputs: the **innovative effort of labour** and the existing level of technology



Direct output: new insights and ideas



Ultimate output: increases $(A_{t+1} - A_t)$ in A_t

IDEAS AS ECONOMIC GOODS

- Ideas are like **public goods**: non-rival and imperfectly excludable.
 - Non-rival: The consumption or use of the good by one person does not prevent other persons from using it. As an example, consider calculus.
- Thus the cost of an idea is fixed (better: **sunk**) cost in many cases.
 - Either it can be bought from an independent R&D firm or developed in an internal company research department.
 - But once acquired, the use of the idea itself is costless.

- In order to have a private production of idea-based technological progress, it must somehow be ensured that the inventor of the idea holds monopoly power over the use of the idea.
- However, if one ensures a private production of ideas by giving monopoly in the use of ideas to the creators, then the **resulting ideas will be used too little** as compared to a social optimum.

PUBLIC PRODUCTION OF IDEAS

- The preceding analysis could suggest that R&D should be conducted by public or other non-profit enterprises.
- However, the costs of development would have to be financed by **distortionary taxes**.
- Furthermore, the general public may wish to have private R&D activities, and be willing to live with the monopoly power necessarily associated with it, in so far as the profit motive creates strong incentives for technological development.

EXCLUDABILITY AND PATENTS

- A good is excludable if the owner of the good can exclude other persons or firms from using it.
- While non-rivalry prevents private idea-based production under perfect competition, complete non-excludability prevents private production under any market structure.
- A patent gives a firm the right to be the sole user of an idea for some period.
- Thus patents work by ensuring both a large degree of excludability and the monopoly power required for a private production of ideas

SUMMING UP

- The microeconomics associated with R&D is subtle: private production of ideas requires imperfect competition and arrangements to ensure excludability such as patents.
- Public production with open access **could** be well motivated from a welfare viewpoint, but we may want to use the profit motive as an engine for technological progress.
- For now we disregard microeconomic subtleties and formulate a more traditional macro model: a set of growth and technology relations – including R&D.

AN R&D ROMER MACROMODEL OF ENDOGENOUS GROWTH

- Aims at explaining the relatively constant growth in GDP per capita over 100-200 years in the “developed world”
- Considers the developed world as one big economy producing all new technology itself
- Two types of output: **final goods** and **new technology**

The final goods sector

$$Y_t = K_t^\alpha (A_t L_{Yt})^{1-\alpha}, \quad 0 < \alpha < 1.$$

- Inputs: K_t , L_{Yt} and A_t where A_t is now considered as the productive effect of the stock of all previous ideas.
- The individual firm can decide its use of capital and labour but takes technology, A_t , as given. All firms can use A_t simultaneously because ideas are non-rival.
- Constant returns to the rival inputs K_t and L_{Yt} : the replication argument. Increasing returns⁹ to K_t , L_{Yt} **and** A_t .

THE R&D SECTOR

- One representative type of firms produce the sector's aggregate output of new technology, $A_{t+1} - A_t$.
- However, each firm is considered as small relative to the entire R&D sector and to the economy.
- It therefore takes total the labour input, L_{At} , the aggregate production, $A_{t+1} - A_t$, and the aggregate stock of knowledge, A_t , as given.
- The individual firm's production function is:

$$a_t = \bar{\rho} A_t^\phi L_{At}^d, \quad \bar{\rho} > 0,$$

where a_t is the number of produced ideas and L_{At}^d is labour input at the **firm level**.

- At the **macro level**, $a_t = A_{t+1} - A_t$ and $L_{At}^d = L_{At}$. No capital input.

- The stock of knowledge, A_t , has an output elasticity, ϕ , of its own:
 - Existing technology can be productive in the production of new technology just as it is in the production of final goods: "standing on shoulders". Speaks for $\phi > 0$, possibly $\phi = 1$.
 - Could have a counter-productive effect: "fishing out". This speaks for $\phi < 0$.
 - We will always assume $\phi > 0$. Fishing out could perhaps motivate $\phi < 1$. In Romer's original contribution, $\phi = 1$.
- There is a possibility of a negative external effect of aggregate labour input in the R&D sector due to "stepping on toes":

$$\bar{\rho} = \rho L_{At}^{\lambda-1}, \quad \rho > 0, \quad 0 < \lambda \leq 1.$$

This effect is absent if $\lambda = 1$.

Summing up:

- At **individual firm level**:

$$a_t = \bar{\rho} A_t^\phi L_{At}^d, \quad \bar{\rho} > 0$$

- At **aggregate level**:

$$\bar{\rho} = \rho L_{At}^{\lambda-1}, \quad a_t = A_{t+1} - A_t \quad \text{and} \quad L_{At}^d = L_{At}$$

- Combining these equations gives the aggregate production function of the R&D sector:

$$A_{t+1} - A_t = \rho A_t^\phi L_{At}^\lambda$$

Special cases:

- $\lambda = 1$ implies a proportional relationship between $A_{t+1} - A_t$ and L_{At} , that is, constant returns to L_{At} .
- $\phi = 1$ implies constant returns to A_t , and the growth rate of technology:

$$\frac{A_{t+1} - A_t}{A_t} = \rho L_{At}^\lambda.$$

THE COMPLETE MODEL

$$Y_t = K_t^\alpha (A_t L_{Yt})^{1-\alpha}$$

$$A_{t+1} - A_t = \rho A_t^\phi L_{At}^\lambda$$

$$K_{t+1} = s Y_t + (1 - \delta) K_t$$

$$L_{t+1} = (1 + n) L_t$$

$$L_{Yt} + L_{At} = L_t$$

$$L_{At} = s_R L_t.$$

Parameters: $\alpha, \phi, \lambda, s, \delta, n$ and s_R

Note: s_R is the R&D share, an exogenous constant, $0 < s_R < 1$.

Labour used in one sector cannot be used in the other.

State variables: K_t , L_t and A_t .

THE LAW OF MOTION

The growth rate of technology

$$g_t \equiv \frac{A_{t+1} - A_t}{A_t} = \rho A_t^{\phi-1} L_{At}^\lambda = \rho A_t^{\phi-1} (s_R L_t)^\lambda \Rightarrow$$

(assuming that ρ and s_R are constant)

$$\frac{g_{t+1}}{g_t} \equiv \left(\frac{A_{t+1}}{A_t} \right)^{\phi-1} \left(\frac{L_{t+1}}{L_t} \right)^\lambda = (1 + g_t)^{\phi-1} (1 + n)^\lambda \Leftrightarrow$$
$$g_{t+1} = (1 + n)^\lambda g_t (1 + g_t)^{\phi-1}. \quad (*)$$

The technological growth rate, g_t , evolves according to its own dynamic equation.

The economic variables

- Define

$$\tilde{k}_t \equiv \frac{K_t}{A_t L_t} = \frac{k_t}{A_t} \quad \text{and} \quad \tilde{y}_t \equiv \frac{Y_t}{A_t L_t} = \frac{y_t}{A_t}.$$

- Then:

$$Y_t = K_t^\alpha (A_t L_{Yt})^{1-\alpha} = K_t^\alpha (A_t (1-s_R) L_t)^{1-\alpha} \Leftrightarrow \tilde{y}_t = \tilde{k}_t^\alpha (1-s_R)^{1-\alpha}.$$

- Dividing on both sides of $K_{t+1} = sY_t + (1-\delta)K_t$ by $A_{t+1}L_{t+1}$ gives

$$\tilde{k}_{t+1} = \frac{1}{(1+n)(1+g_t)} \left(s\tilde{k}_t^\alpha (1-s_R)^{1-\alpha} + (1-\delta)\tilde{k}_t \right). \quad (**)$$

The law of motion

consists of (*) and (**), a system of two, one-way coupled, difference equations in g_t and $\tilde{k}_t$.

The independent dynamics of technology:

1. $A_{t+1} - A_t = \rho A_t^\phi L_{At}^\lambda$ is strictly positive (as long as $L_{At} > 0$) implying that $g_t > 0$. **Knowledge is accumulated all the time.**

In the following, consider the case of a constant labour input in the R&D sector, $L_{At} = L_A \Rightarrow$

$$A_{t+1} - A_t = \rho A_t^\phi L_A^\lambda \text{ and } g_t = \rho A_t^{\phi-1} L_A^\lambda.$$

2. If $\phi = 1$, then $g_t = \rho L_A^\lambda$ is constant, implying that $A_{t+1} - A_t$ is growing and that $A_t \rightarrow \infty$ for $t \rightarrow \infty$. A constant labour input in the R&D sector will maintain a constant growth rate in A_t . Leads to **(strictly) endogenous growth**.
3. If $0 < \phi < 1$, then $A_{t+1} - A_t$ is (again) growing and $A_t \rightarrow \infty$ for $t \rightarrow \infty$, but $g_t \rightarrow 0$ as $t \rightarrow \infty$, because $A_t^{\phi-1} \rightarrow 0$. A constant labour input in R&D cannot maintain a constant growth rate in A_t . For this a growing labour input will be required. Leads to **semi-endogenous growth**.

SEMI-ENDOGENOUS GROWTH

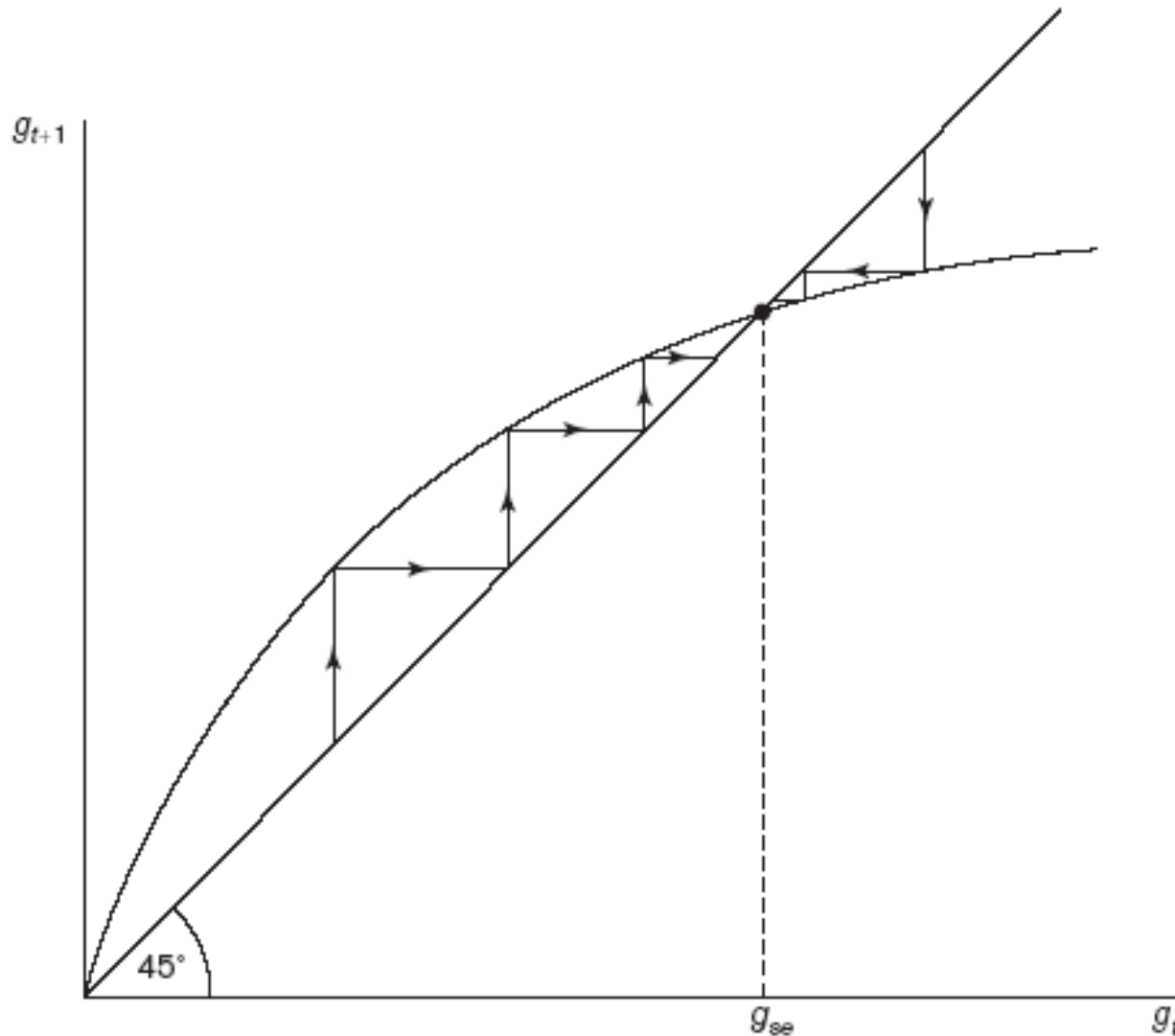
- We assume that $0 < \phi < 1$ and $n > 0$. The dynamic system is:

$$g_{t+1} = (1+n)^\lambda g_t (1+g_t)^{\phi-1},$$

$$\tilde{k}_{t+1} = \frac{1}{(1+n)(1+g_t)} \left(s \tilde{k}_t^\alpha (1-s_R)^{1-\alpha} + (1-\delta) \tilde{k}_t \right).$$

- Consider first the independent law of motion for g_t . Remember that $g_t > 0$. The transition curve for g_t :
 1. Passes through (0,0).
 2. Has slope $(1+n)^\lambda (1+g_t)^{\phi-2} (1+\phi g_t) > 0$.
 3. Has a unique, positive intersection with the 45°-line. Inserting $g_{t+1} = g_t$ gives:
$$g_t = (1+n)^{\frac{\lambda}{1-\phi}} - 1 \equiv g_{se} > 0.$$
 4. The slope in (0,0) is larger than one. (In 3. and 4. we use that $n > 0$).

- This implies that the transition curve for g_t looks as follows:



- In the long run, the technological growth rate, g_t , converges towards $g_{se} = (1 + n)^{\frac{\lambda}{1-\phi}} - 1$.

- Consider now the other dynamic equation:

$$\tilde{k}_{t+1} = \frac{1}{(1+n)(1+g_t)} \left(s\tilde{k}_t^\alpha (1-s_R)^{1-\alpha} + (1-\delta)\tilde{k}_t \right).$$

- Set g_t at its long-run value g_{se} :

$$\tilde{k}_{t+1} = \frac{1}{(1+n)(1+g_{se})} \left(s\tilde{k}_t^\alpha (1-s_R)^{1-\alpha} + (1-\delta)\tilde{k}_t \right).$$

- Similar to the transition equation of the Solow model: convergence towards the constant solution,

$$\tilde{k}^* = \left(\frac{s}{n + g_{se} + \delta + ng_{se}} \right)^{\frac{1}{1-\alpha}} (1-s_R)$$

and hence convergence of $\tilde{y}_t$ to $\tilde{y}^* = \left(\tilde{k}^* \right)^\alpha (1-s_R)^{1-\alpha}$:

$$\tilde{y}^* = \left(\frac{s}{n + g_{se} + \delta + ng_{se}} \right)^{\frac{\alpha}{1-\alpha}} (1-s_R)$$

- Convergence of g_t to g_{se} and of $\tilde{y}_t$ to $\tilde{y}^*$: steady state.

GROWTH IN STEADY STATE

- In steady state, $\tilde{k}_t = k_t / A_t$ and $\tilde{y}_t = y_t / A_t$ are constant. Hence k_t and y_t must grow at the same rate as A_t . In steady state this rate is g_{se} :
- **Capital per worker, output per worker and technology converge towards being growing at one and the same constant rate:** $g_{se} = (1 + n)^{\lambda / (1 - \phi)} - 1$. We have:
 - Balanced growth in the long run
 - Only **semi**-endogenous growth
 - Structural policy: g_{se} does not depend on s_R or on s , only on n (plus λ, ϕ)
 - Empirics: recall cross country and panel data from Chapter 8
- If the empirics do not seem to support semi-endogenous growth, we do not have to reject the model – only its steady state.
- Therefore: look at very slow convergence as occurring for $\phi \rightarrow 1$.

STRICTLY ENDOGENOUS GROWTH

- We assume that $\phi = 1$ and $n = 0$.
- The equation for population growth, $L_{t+1} = (1 + n)L_t$, becomes $L_t = L$, while the equation describing technological growth, $g_t = \rho A_t^{\phi-1} (s_R L_t)^\lambda$, becomes:

$$g_t = \rho (s_R L)^\lambda \equiv g_e$$

That is, we have a constant technological growth rate.

- The model reduces to:

$$Y_t = K_t^\alpha \left(A_t (1 - s_R) L \right)^{1-\alpha}$$

$$K_{t+1} = s Y_t + (1 - \delta) K_t$$

$$A_{t+1} = (1 + g_e) A_t.$$

- This is like an ordinary Solow model (of exogenous growth), with no population growth, technological growth rate g_e , and labour force $(1 - s_R) L$.

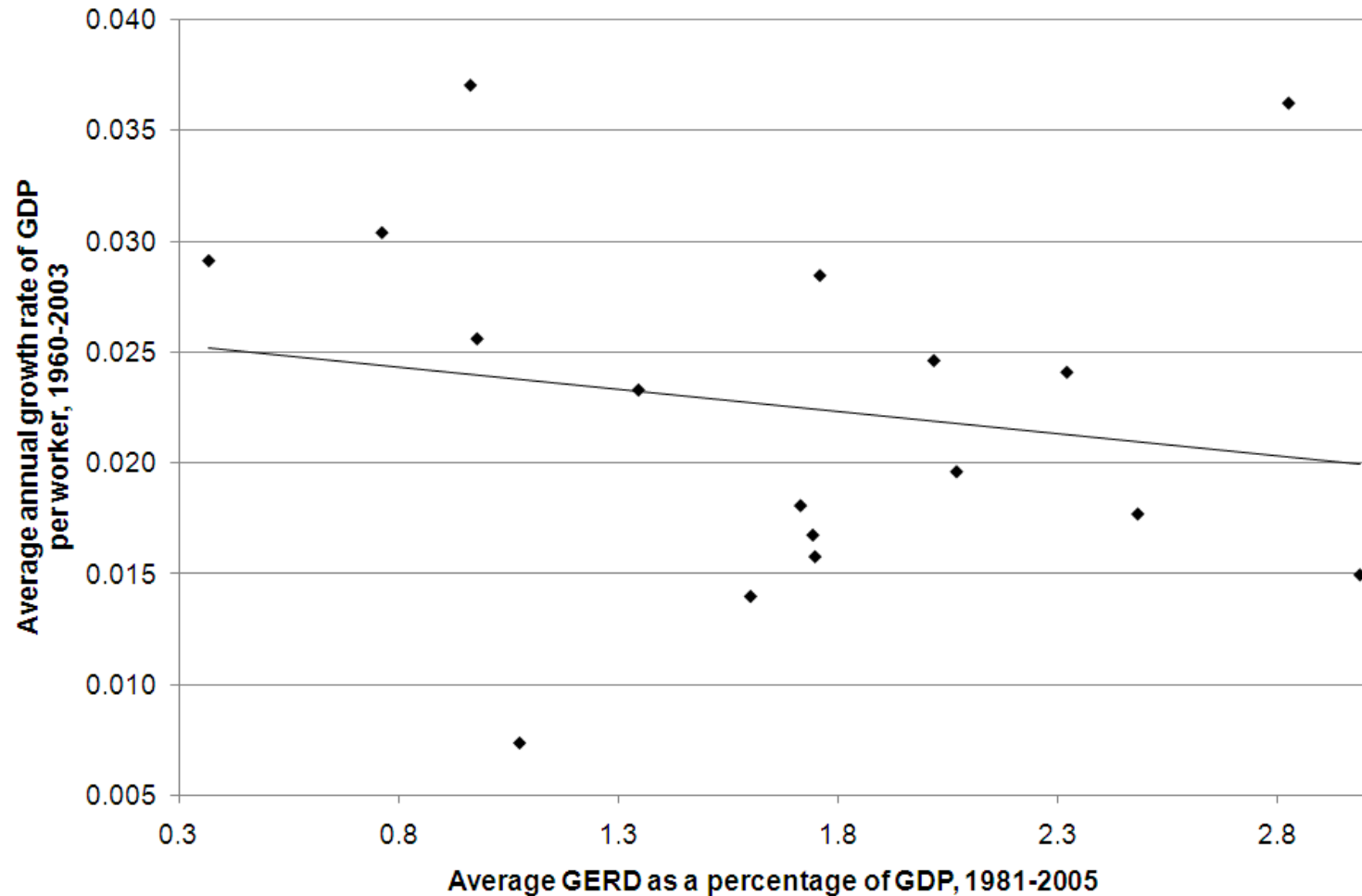
- From a usual analysis, $\tilde{k}_t = k_t / A_t$ and $\tilde{y}_t = y_t / A_t$ will converge towards constant steady state values:
- **Technology, A_t , grows at a constant rate**

$$g_t = \rho(s_R L)^\lambda \equiv g_e$$

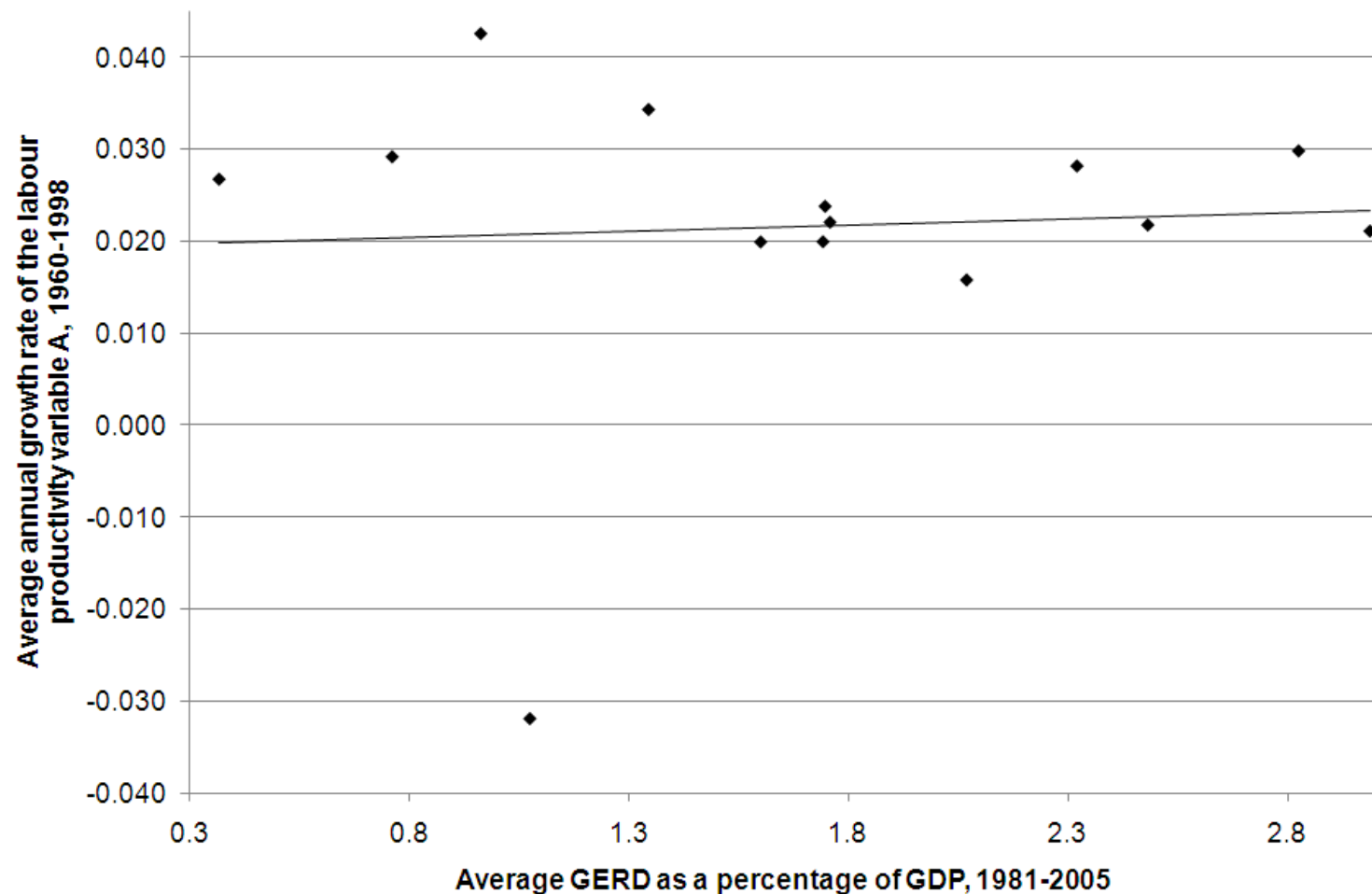
while capital per worker, k_t , and GDP per worker, y_t , converge towards growing at the rate of g_e .

- Some good features of the model:
 - Balanced growth in the long run
 - Strictly endogenous growth
 - Structural policy implication: g_{se} depends on s_R
 - the model has convergence
- Empirics:

Average annual growth rate of real GDP per worker against GERD as a fraction of GDP, 17 OECD countries



Average annual rate of labour-augmenting progress against average GERD as a fraction of GDP, 17 OECD countries



Source: OECD Science, Technology and Industry Outlook, 2002; and data set for Bernanke and Gurkaynak (2001).

- Some problematic features of the model:
 - g_{se} is independent of s . Problem? No. In this macro model s_R is exogenous, but in a model where s_R is endogenous it will depend positively on s .
 - Does s_R affect the growth rate positively across countries? No! Is that a problem? Not if the model is not a country by country model, but for the world.
 - A larger L implies a larger long-run growth rate, g_e . Is this a problem? **Yes, this is a serious and problematic scale effect!**

CONCLUDING REMARKS

1. Semi-endogenous growth may seem implausible, because the empirics do not support the predicted positive correlation between economic growth and growth of population (labour force). However, remember the “global, long-run argument” we presented in connection with Chapter 8.
2. Strictly endogenous growth may seem implausible because of the scale effect.

A hybrid: **“hemi-endogenous” growth**

3. Assumes $\phi < 1$, ruling out scale effects. In hemi-endogenous growth it is assumed that the needed increase in labour input, $L_{At} = s_R L_t$, in R&D does not come from increases in L_t , but from slow and gradual increases in s_R . It hinges on the following property of the model: a small increase in s_R gives a very long adjustment for reasonable parameter values. If this is the correct explanation of the last 200 years of growth then **one day the economic growth rate will tend to zero.**